Effects of sugar-sweetened and sugar-free cocoa on endothelial function in overweight adults.

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BACKGROUND: Studies of cocoa suggest an array of cardiovascular benefits; however, the effects of daily intake of sugar-free and sugar-sweetened cocoa beverages on endothelial function (EF) have yet to be established. METHODS: 44 adults (BMI 25-35 kg/m2) participated in a randomized, controlled, crossover trial. Participants were randomly assigned to a treatment sequence: sugar-free cocoa beverage, sugar-sweetened cocoa beverage, and sugar-sweetened cocoa-free placebo. Treatments were administered daily for 6 weeks, with a 4-week washout period. RESULTS: Cocoa ingestion improved EF measured as flow-mediated dilation (FMD) compared to placebo (sugar-free cocoa: change, 2.4% [95% CI, 1.5 to 3.2] vs. -0.8% [95% CI, -1.9 to 0.3]; difference, 3.2% [95% CI, 1.8 to 4.6]; p<0.001 and sugar-sweetened cocoa: change, 1.5% [95% CI, 0.6 to 2.4] vs. -0.8% [95% CI, -1.9 to 0.3]; difference, 2.3% [95% CI, 0.9 to 3.7]; p=0.002). The magnitude of improvement in FMD after consumption of sugar-free versus sugar-sweetened cocoa was greater, but not significantly. Other biomarkers of cardiac risk did not change appreciably from baseline. BMI remained stable throughout the study. CONCLUSIONS: Daily cocoa ingestion improves EF independently of other biomarkers of cardiac risk, and does not cause weight gain. Sugar-free preparations may further augment endothelial function.